Secondary Structure Prediction using PDB Protein Structures

1.Student Details

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2. Objective

The objective of this project is to identify and predict protein secondary structures from PDB protein structure files. The project identifies important protein secondary structures such as **alpha helices**, **beta sheets**, and **coil/loop regions** from amino acid residue information.

3. Introduction

Proteins are large biological molecules made up of amino acids. The sequence and arrangement of amino acids determine the shape and function of a protein. Protein structures are usually divided into different levels such as primary structure, secondary structure, tertiary structure, and quaternary structure.

The secondary structure of a protein mainly contains **alpha helices**, **beta sheets**, and **coil/loop regions**..

4. Database Used

The database used in this project is the **Protein Data Bank**, also called **PDB**.

The Protein Data Bank contains three-dimensional structural data of proteins, nucleic acids, and biological molecules. PDB files contain information about amino acid residues, atom positions, chains, and secondary structure records such as HELIX and SHEET.

For this project, the protein structure file used is:

1CRN.pdb

5. Tools Used

The tools used in this project are:

Python

Python is used to write the main program for reading and processing the PDB file.

Biopython

Biopython is used to handle biological data and parse protein structure files.

Scikit-learn

Scikit-learn is used to create a basic machine learning prediction model.

PyMOL

PyMOL is used to visualize the protein structure in cartoon form.

AWK-style Parsing

AWK-style parsing is used to extract only important records like HELIX and SHEET from the PDB file.

6. System Requirements

Operating System: Windows
Programming Language: Python
Libraries: Biopython, Scikit-learn
Visualization Tool: PyMOL
Input File: 1CRN.pdb

7. Project Folder Structure

The project folder is arranged as follows:

Name	Status	Date modified	Type	Size
pdb_files		16-05-2026 01:43	File folder	
awk_parse.py		16-05-2026 02:38	Python Source File	1 KB
main.py		16-05-2026 02:36	Python Source File	3 KB
ml_model.py		16-05-2026 02:38	Python Source File	3 KB

8. Methodology

The project is completed in the following steps:

Step 1: Download PDB file from Protein Data Bank.
Step 2: Store the PDB file inside the project folder.
Step 3: Install Python libraries such as Biopython and Scikit-learn.
Step 4: Read the PDB file using Python.
Step 5: Extract HELIX and SHEET records.
Step 6: Extract amino acid residue information.
Step 7: Classify residues as Alpha Helix, Beta Sheet, or Coil/Loop.
Step 8: Display the output in table format.
Step 9: Display terminal graph.
Step 10: Perform AWK-style parsing.
Step 11: Train a basic machine learning model.
Step 12: Visualize the protein structure using PyMOL.

9. Process / Implementation

Step 1: Download PDB File

The PDB file was downloaded from the Protein Data Bank website. The protein ID used in this project is:

1CRN

The downloaded file was saved as:

1CRN.pdb

Step 2: Install Required Libraries

The following command was used to install Biopython and Scikit-learn:

pip install biopython scikit-learn

If needed, this command can also be used:

python -m pip install biopython scikit-learn

Step 3: Main Python Program

The main Python file is:

main.py

This program reads the PDB file, extracts amino acid residues, identifies alpha helices and beta sheets, and displays the output in the terminal.

Step 4: Run Main Program

The program was executed using the command:

python main.py

The output shows residue number, amino acid name, and secondary structure type.

Output:

```
PS C:\Users\Dheeraj\OneDrive\Desktop\SecondaryStructureProject> python main.py

SECONDARY STRUCTURE PREDICTION USING PDB
----------------------------------------

Chain: A
Residue No       Amino Acid       Structure
------------------------------------------
1                THR              Beta Sheet
2                THR              Beta Sheet
3                CYS              Beta Sheet
4                CYS              Beta Sheet
5                PRO              Coil / Loop
6                SER              Coil / Loop
7                ILE              Alpha Helix
8                VAL              Alpha Helix
9                ALA              Alpha Helix
10               ARG              Alpha Helix
11               SER              Alpha Helix
12               ASN              Alpha Helix
13               PHE              Alpha Helix
14               ASN              Alpha Helix
15               VAL              Alpha Helix
16               CYS              Alpha Helix
17               ARG              Alpha Helix
18               LEU              Alpha Helix
19               PRO              Alpha Helix
20               GLY              Coil / Loop
```

Step 5: Terminal Graph

The program also displays a terminal graph showing the distribution of secondary structures.

The graph contains:

Alpha Helix
Beta Sheet
Coil / Loop

TERMINAL GRAPH

```
TERMINAL GRAPH
----------------------------------------
Alpha Helix     |                                     21 residues (45.65%)
Beta Sheet      |                                     8 residues (17.39%)
Coil / Loop     |                                     17 residues (36.96%)

SUMMARY
----------------------------------------
Total Residues : 46
Alpha Helix    : 21
Beta Sheet     : 8
Coil / Loop    : 17
```

Step 6: AWK-style Parsing

The file used for AWK-style parsing is:

awk_parse.py

This file reads the PDB file and extracts only HELIX and SHEET records.

The command used is:

python awk_parse.py

This completes the scripting part of the project.

Since Windows does not directly support AWK by default, the same AWK-style text parsing was implemented using Python.

Output:

```
PS C:\Users\Dheeraj\OneDrive\Desktop\SecondaryStructureProject> python awk_parse.py
AWK STYLE PDB PARSING
---------------------
Reading only HELIX and SHEET records from PDB file

HELIX RECORD FOUND:
HELIX    1  H1 ILE A    7  PRO A   19  13/10 CONFORMATION RES 17,19       13
HELIX RECORD FOUND:
HELIX    2  H2 GLU A   23  THR A   30  1DISTORTED 3/10 AT RES 30           8
SHEET RECORD FOUND:
SHEET    1  S1 2 THR A   1  CYS A   4  0
SHEET RECORD FOUND:
SHEET    2  S1 2 CYS A  32  ILE A  35 -1

AWK-style parsing completed.
```

Step 7: Machine Learning Model

The file used for machine learning is:

ml_model.py

This program creates a basic prediction model using amino acid codes as input and secondary structure labels as output.

The command used is:

python ml_model.py

The machine learning model used is:

Random Forest Classifier

The model predicts whether an amino acid residue belongs to:

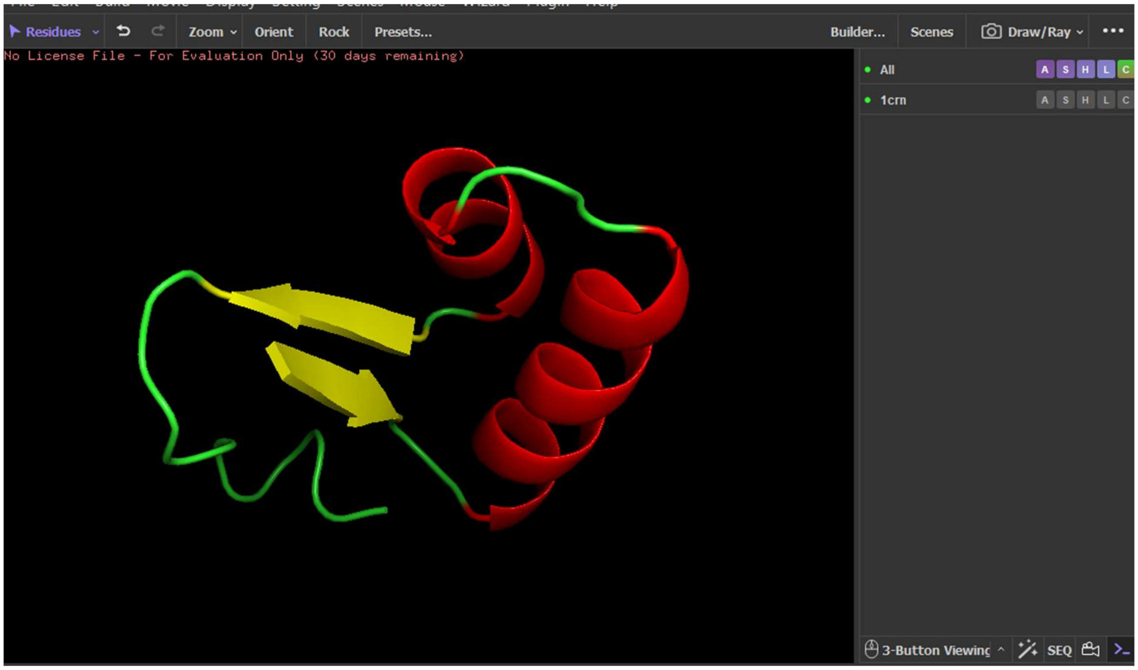
Alpha Helix
Beta Sheet
Coil / Loop

Output:

```
PS C:\Users\Dheeraj\OneDrive\Desktop\SecondaryStructureProject> python ml_model.py
MACHINE LEARNING MODEL
----------------------
Dataset created from PDB file
Total samples: 46

Model training completed.

Classification Report:
              precision    recall  f1-score   support

 Alpha Helix       0.67      0.33      0.44         6
  Beta Sheet       0.33      1.00      0.50         2
 Coil / Loop       0.80      0.67      0.73         6

    accuracy                           0.57        14
   macro avg       0.60      0.67      0.56        14
weighted avg       0.68      0.57      0.57        14


Example Prediction
------------------
Amino acid: CYS
Predicted secondary structure: Beta Sheet
```

Step 8: PyMOL Visualization

PyMOL was used to visualize the protein structure.



PyMOL gives a clear visual representation of the secondary structure of the protein.

10. Algorithm

Step 1: Start the program.
Step 2: Import required Python libraries.
Step 3: Load the PDB file.
Step 4: Read HELIX records from the PDB file.
Step 5: Read SHEET records from the PDB file.
Step 6: Extract amino acid residues using Biopython.
Step 7: Check each residue number.
Step 8: If the residue is present in HELIX record, classify it as Alpha Helix.
Step 9: If the residue is present in SHEET record, classify it as Beta Sheet.
Step 10: Otherwise classify it as Coil/Loop.
Step 11: Count the number of residues in each structure type.
Step 12: Display the result in terminal.
Step 13: Display terminal graph.
Step 14: Train a basic prediction model.
Step 15: Visualize the protein using PyMOL.
Step 16: Stop the program.

11. Result

The project successfully identified the secondary structures present in the PDB protein file.

The result displayed:

1. Amino acid residue number
2. Amino acid name
3. Secondary structure type
4. Count of alpha helix residues
5. Count of beta sheet residues
6. Count of coil/loop residues
7. Terminal graph
8. Machine learning prediction output
9. PyMOL visualization

12. Applications

Protein secondary structure prediction is useful in:

1. Bioinformatics research
2. Protein structure analysis
3. Drug discovery
4. Disease-related protein study
5. Molecular biology
6. Computational biology
7. Protein folding studies

13. Advantages

1. Easy to understand protein structure.
2. Uses real PDB protein data.
3. Helps identify alpha helices and beta sheets.
4. Uses Python and Biopython for biological data processing.
5. Gives terminal graph output.
6. Includes basic machine learning model.
7. PyMOL gives clear visualization.

14. Limitations

1. The project uses only one PDB file.
2. The basic machine learning model is trained on small data.
3. High accuracy requires many protein structure files.
4. TensorFlow-based CNN and Bidirectional LSTM are not implemented in the final working version.
5. The model depends on existing HELIX and SHEET records in the PDB file.

15. Future Enhancement

In the future, this project can be improved by using more PDB files and larger datasets. Advanced deep learning models such as **CNN** and **Bidirectional LSTM** can be used to predict protein secondary structures directly from amino acid sequences.

CNN can identify local amino acid patterns, while Bidirectional LSTM can learn sequence information from both forward and backward directions. This can improve prediction accuracy when a large protein dataset is available.

16. Conclusion

This project successfully performs secondary structure prediction using PDB protein structure files. The PDB file was downloaded from the Protein Data Bank and processed using Python and Biopython.

The program extracted amino acid residue information and classified residues into alpha helix, beta sheet, and coil/loop regions. A terminal graph was displayed to show the distribution of secondary structures. AWK-style parsing was used to extract important PDB records such as HELIX and SHEET. A basic machine learning model was also trained for prediction. Finally, PyMOL was used to visualize the protein structure.

Thus, the project demonstrates how computational tools can be used to analyze protein secondary structures.